

An aerial photograph of the University of Colorado Boulder campus. The foreground is filled with trees showing vibrant autumn foliage in shades of yellow, orange, and red. In the middle ground, several university buildings are visible, including a prominent red brick building with a tall, ornate clock tower. An American flag flies from a tall pole on the roof of the clock tower. The background features a large, rugged mountain with rocky peaks under a clear blue sky.

Data Infrastructure

CSCI 5942

Lecture 6



College of Engineering & Applied Science

UNIVERSITY OF COLORADO **BOULDER**

Agenda

- Wrap up discussion of Roofline Model
- Today: “Offline” data infrastructure
- Next class: “Online” data infrastructure

Hardware bottlenecks

1,000 H100s $\times$ 30 days $\times$ 40% MFU?

as measured; imagine why?

MFU is model FLOPs utilization: the fraction of the chips' peak rate the run actually reaches.

I/O bottlenecks (Network)

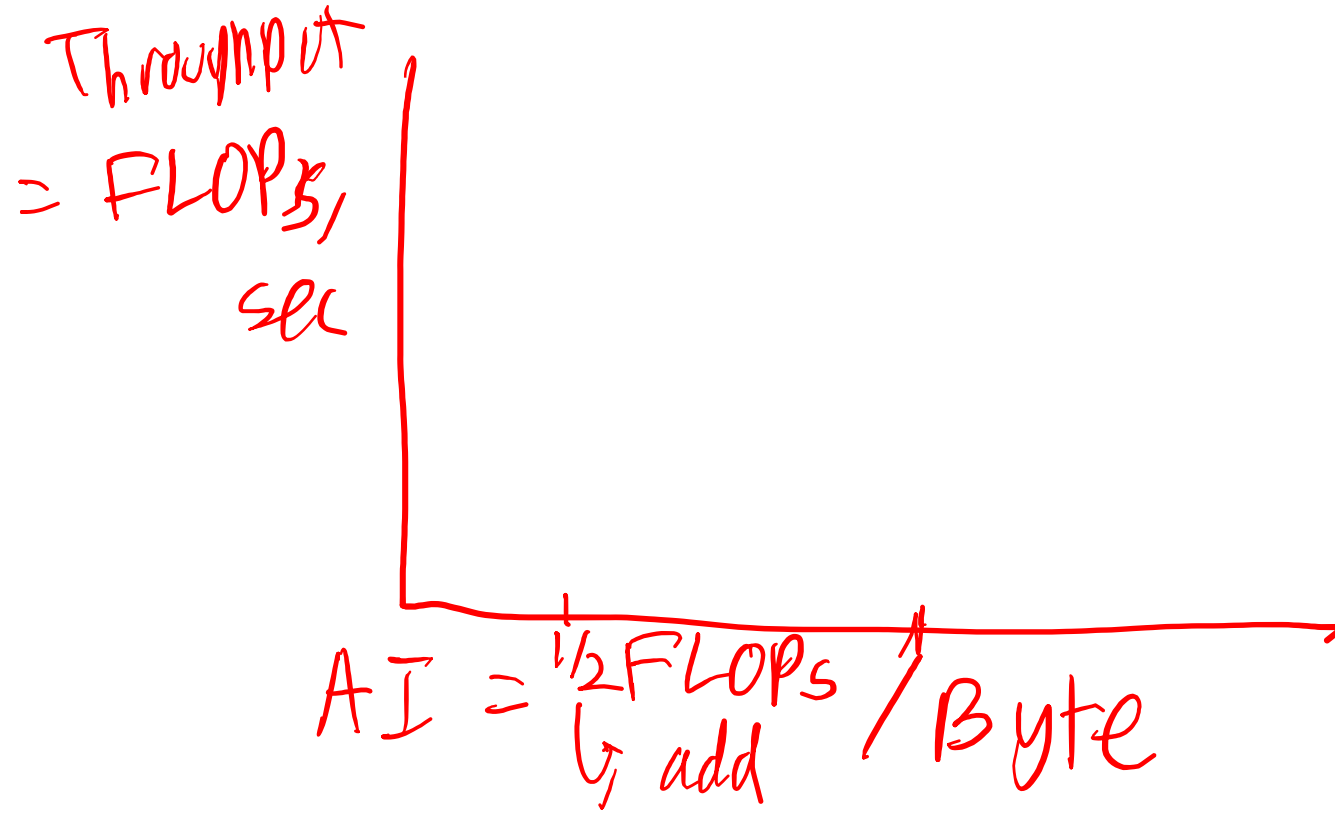


Memory bottlenecks

Hardware bottlenecks

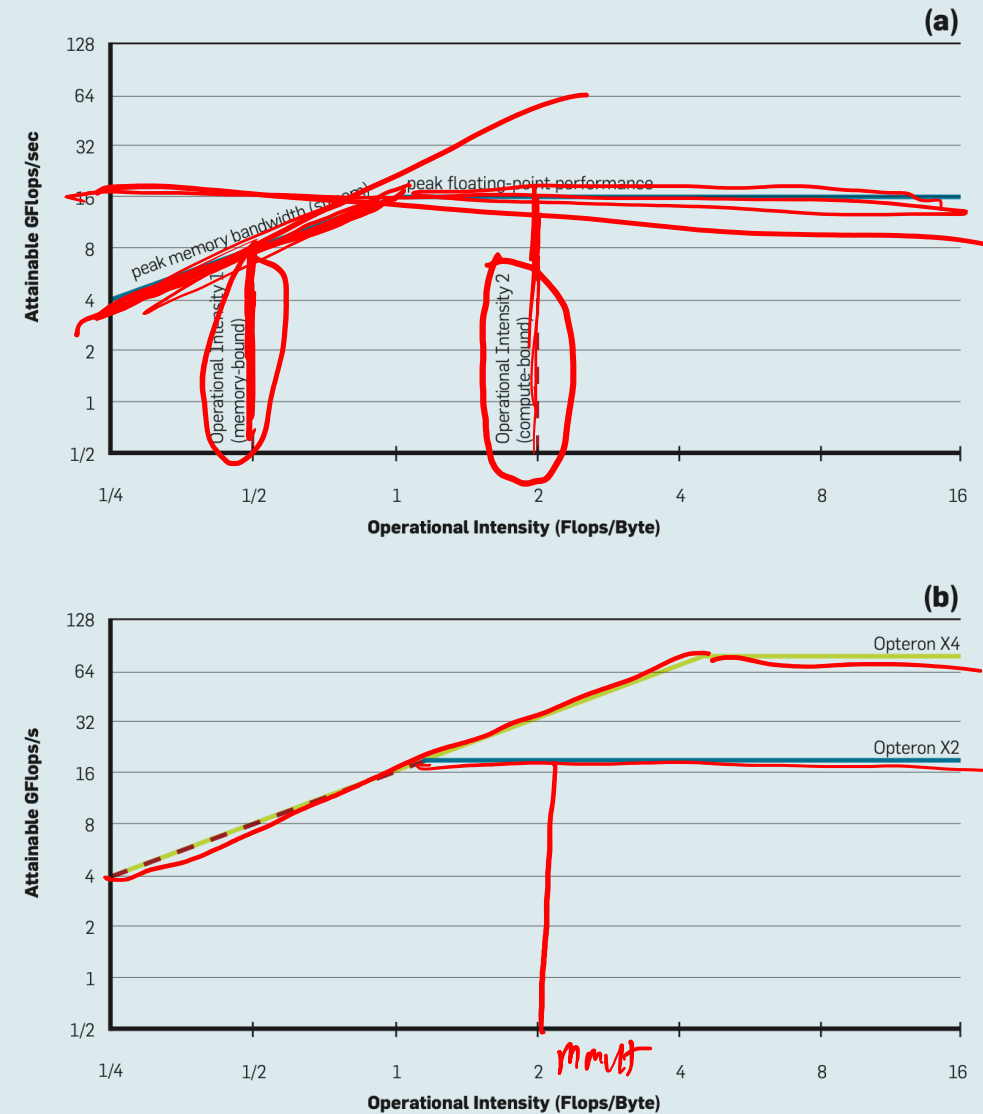
- How can we reason about if a specific model/algorithm/kernel/function is bound by off-chip memory or compute?

Roofline Model



Roofline Model

Figure 1: Roofline model for (a) AMD Opteron X2 and (b) Opteron X2 vs. Opteron X4.

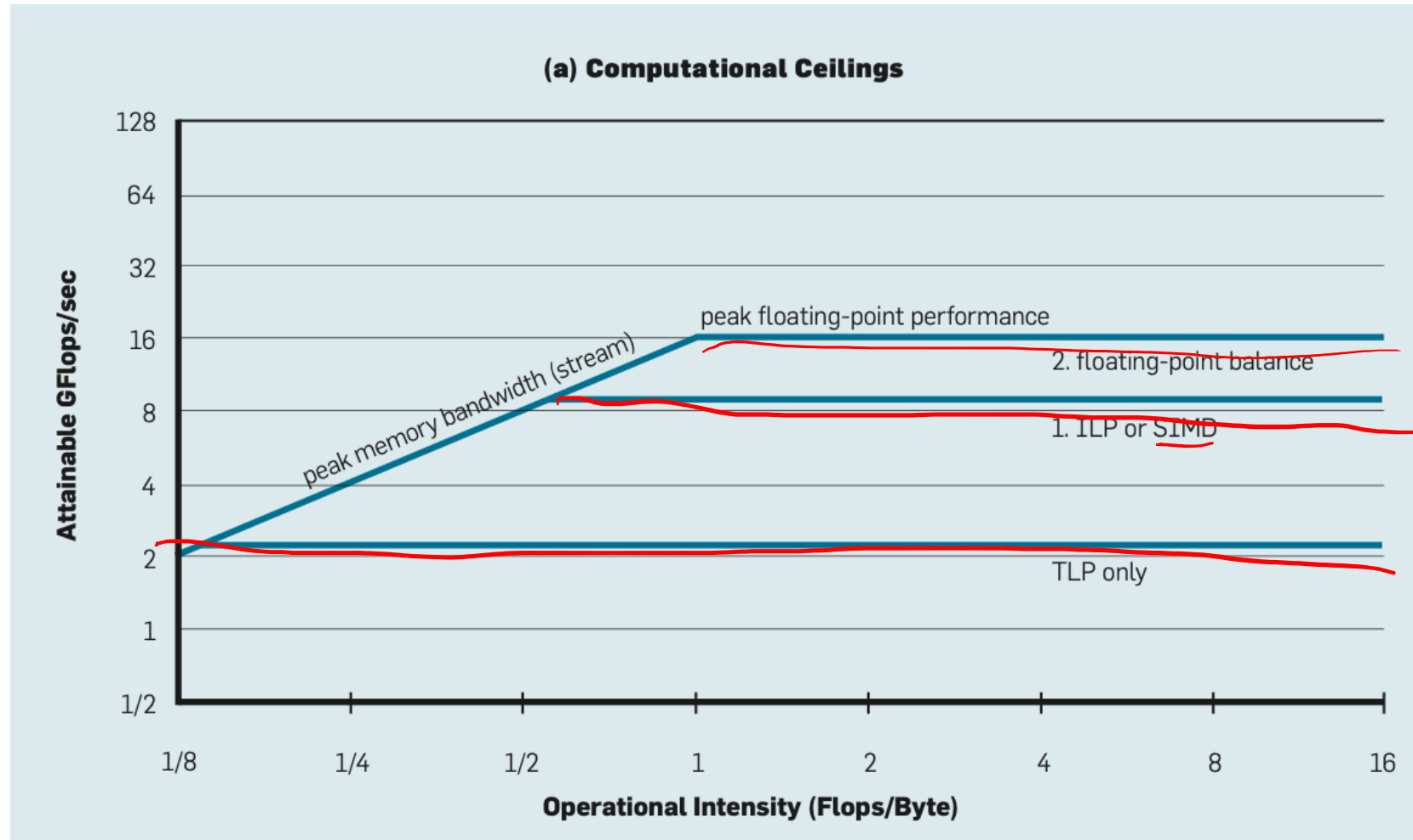


FLOPs/sec
 FLOPs/byte
 $= \text{bytes/sec}$

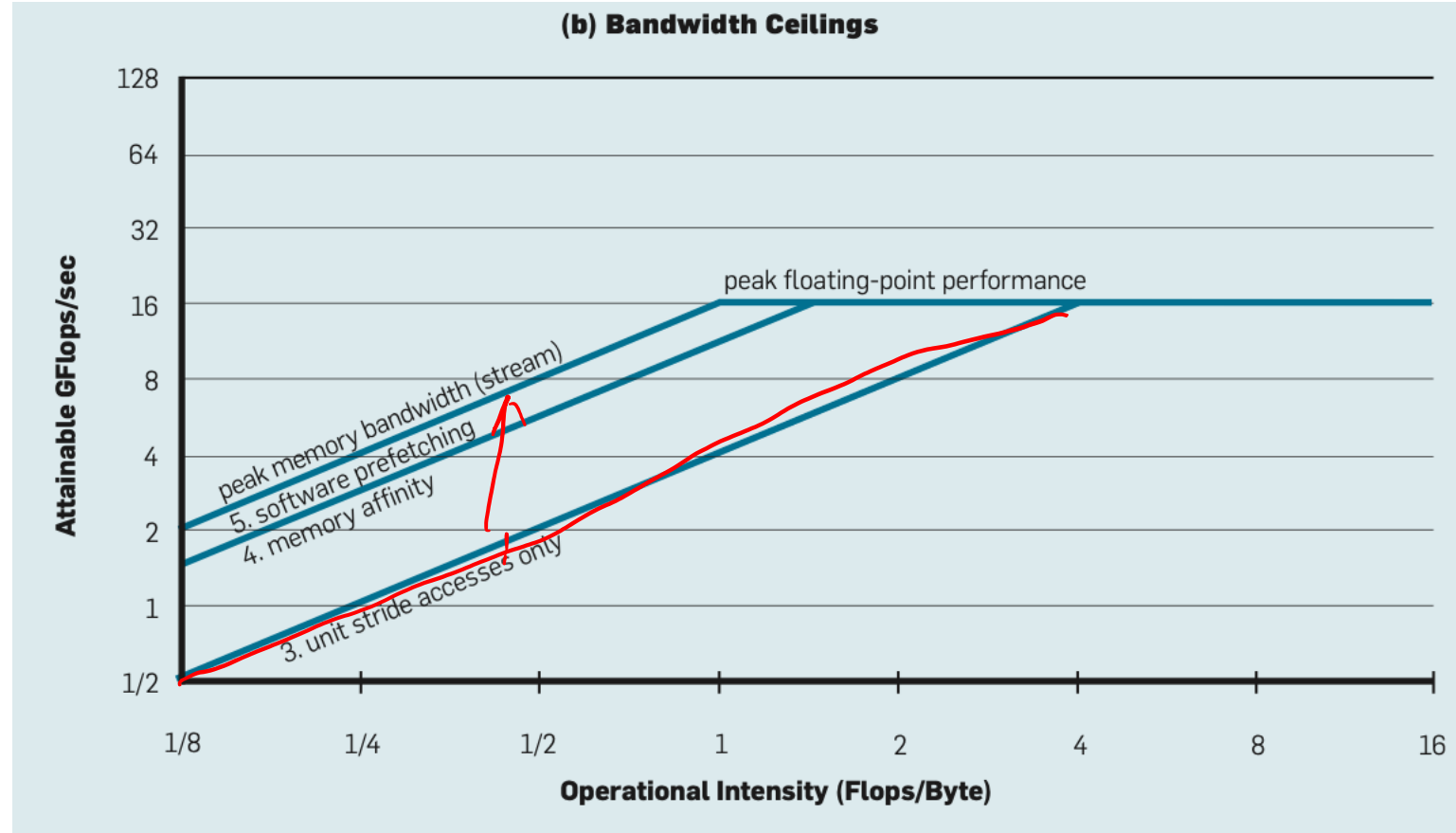
compute

mult

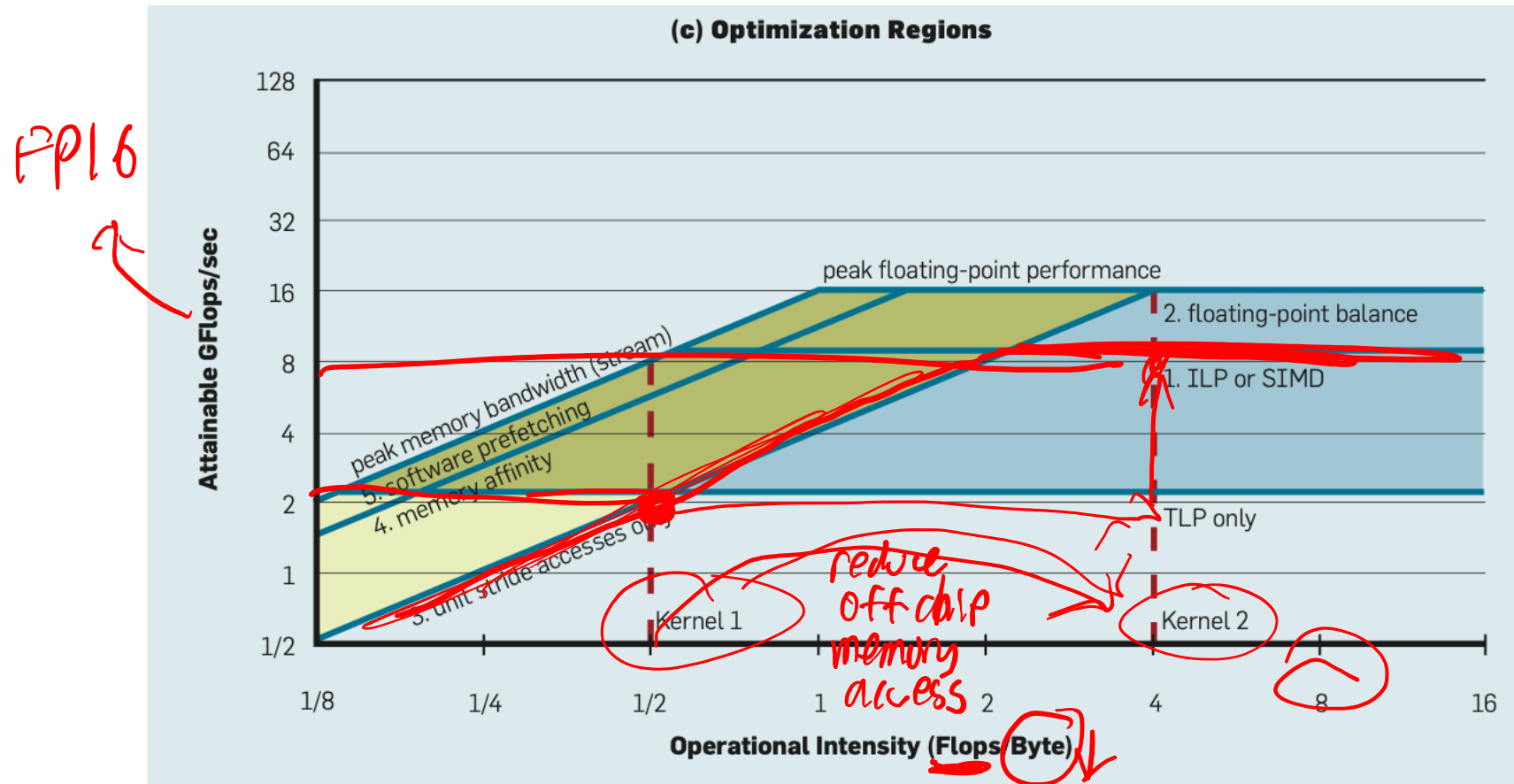
Roofline Model



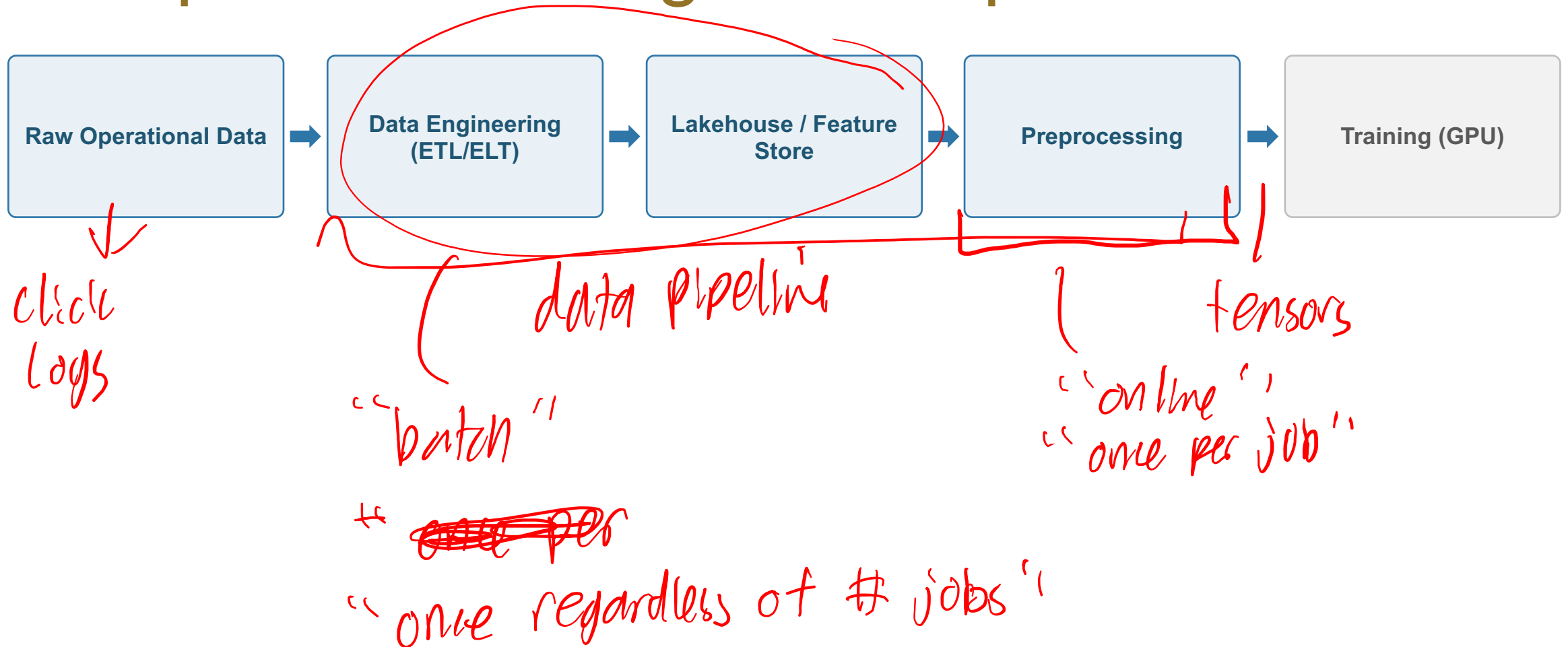
Roofline Model



Roofline Model



Recap: The Training Data Pipeline



Data Engineering

Fits in memory

```
>>> d = {'col1': [1, 2], 'col2': [3, 4]}
>>> df = pd.DataFrame(data=d)
>>> df
```

	col1	col2
0	1	3
1	2	4

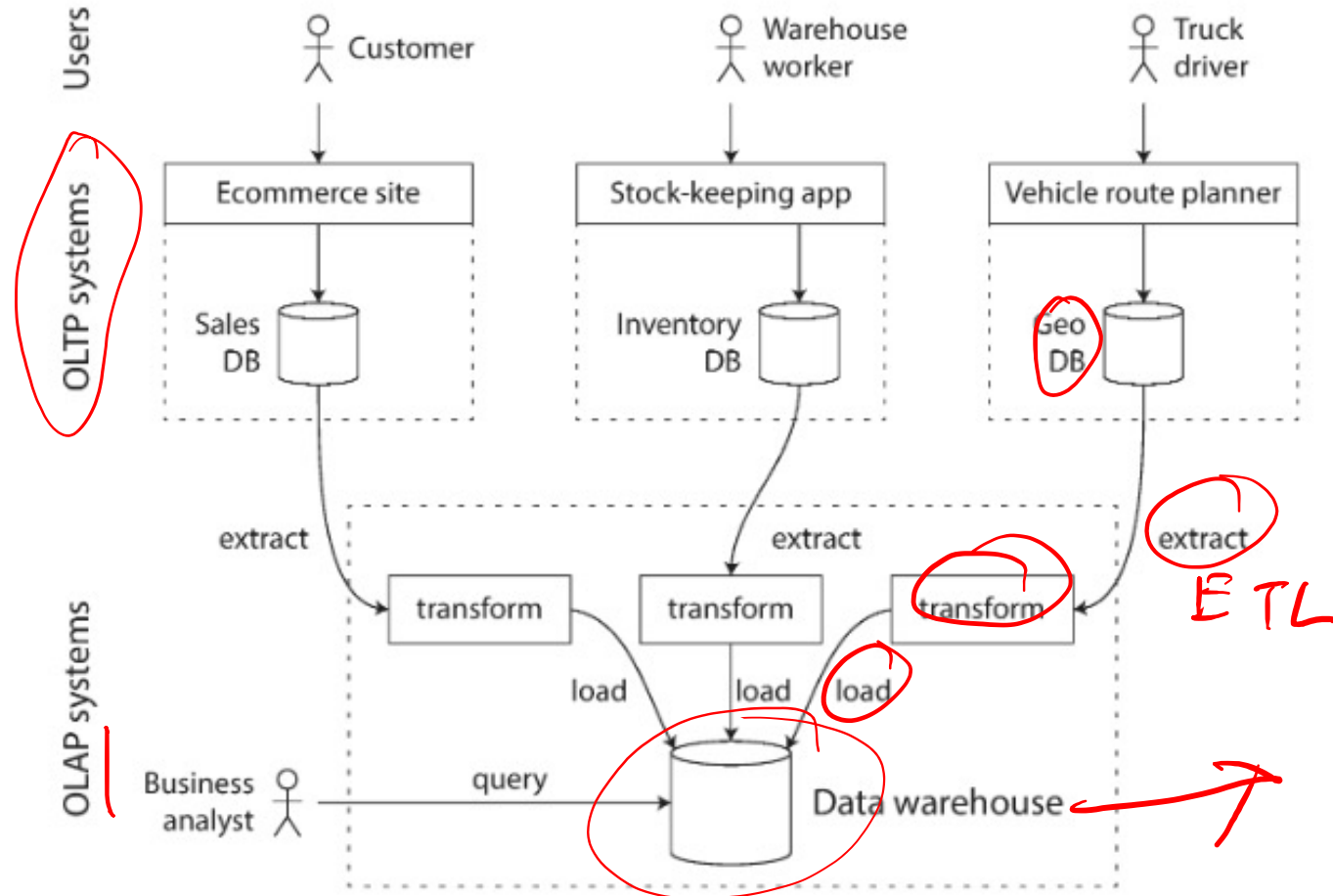
OLTP -> OLAP

- ↳ Transactions
- "daily ops"
- "add \$100 to Bob's acct"

Analytics ~ 2005-2008

- "Business Intelligence"

Data Warehouses ~ 2010s



→ Optimized for large analytic queries

Issues with Data Warehouses

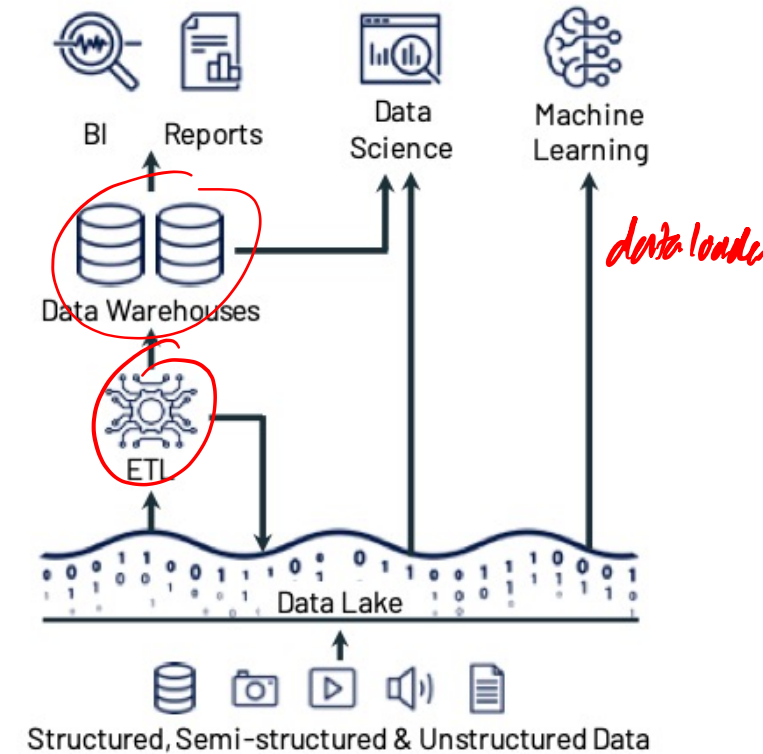
- DW focuses on structured, relational data with a schema-on-write
 - Focused on business intelligence, analytical queries
 - Expensive: Managed by some RDBMS
 - Teradata, Vertica, SAP Hana, RedShift, ...
- What about ML?
 - Lots of unstructured data (videos, images, speech, ...)
 - High volume: can't afford to put everything into warehouse

Data Lake

"Schema-on-read"

- Land raw business data into common, low-cost storage (e.g., object store)
 - Schema-on-Read
 - ETL to data warehouse for BI
- Open file formats and systems
 - Parquet, ORC
 - HDFS, Spark, MapReduce, Hive
 - Cloud: S3, GCS, ADLS

blob store



Issues with Data Lakes

- Consistency/staleness between data lake and downstream data warehouse
- Open formats not as optimized as proprietary storage
- Paying twice for storage (lake + warehouse)
- For ML: lose data warehouse features such as indexing and versioning

Data Lakehouse

- TL;DR: Transactional metadata layer on top of data lake

Lakehouse: A New Generation of Open Platforms that Unify Data Warehousing and Advanced Analytics

Michael Armbrust¹, Ali Ghodsi^{1,2}, Reynold Xin¹, Matei Zaharia^{1,3}

¹Databricks, ²UC Berkeley, ³Stanford University

Abstract

This paper argues that the data warehouse architecture as we know it today will wither in the coming years and be replaced by a new architectural pattern, the Lakehouse, which will (i) be based on open direct-access data formats, such as Apache Parquet, (ii) have first-class support for machine learning and data science, and (iii) offer state-of-the-art performance. Lakehouses can help address several major challenges with data warehouses, including data staleness, reliability, total cost of ownership, data lock-in, and limited use-case support. We discuss how the industry is already moving toward Lakehouses and how this shift may affect work in data management. We also report results from a Lakehouse system using Parquet that is competitive with popular cloud data warehouses on TPC-DS.

1 Introduction

This paper argues that the data warehouse architecture as we know

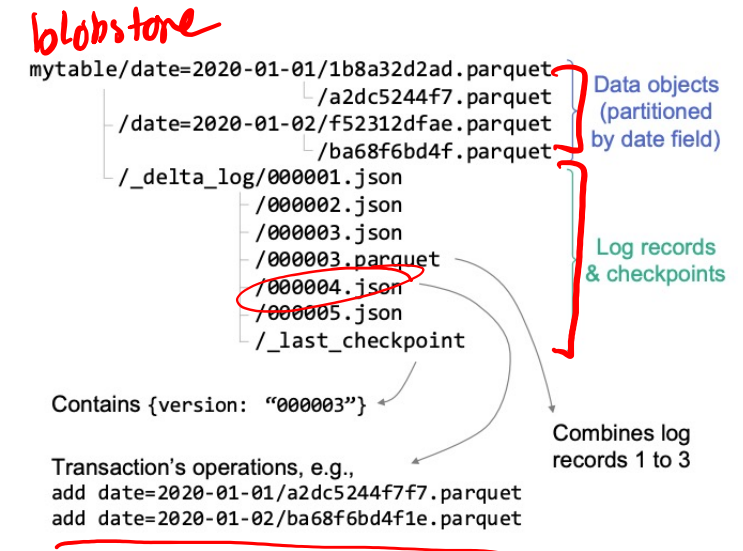
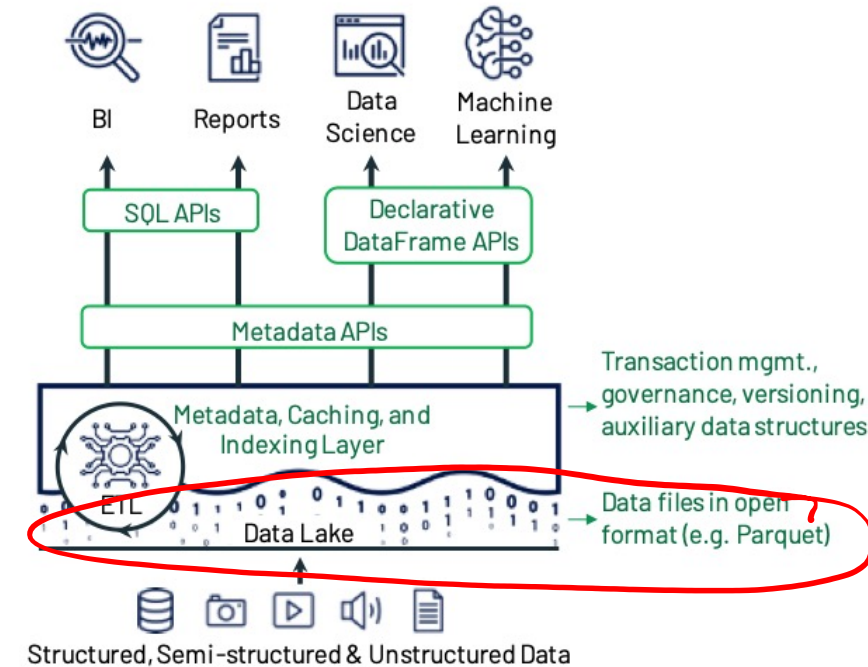
quality and governance downstream. In this architecture, a small subset of data in the lake would later be ETLed to a downstream data warehouse (such as Teradata) for the most important decision support and BI applications. The use of open formats also made data lake data directly accessible to a wide range of other analytics engines, such as machine learning systems [30, 37, 42].

From 2015 onwards, cloud data lakes, such as S3, ADLS and GCS, started replacing HDFS. They have superior durability (often >10 nines), geo-replication, and most importantly, extremely low cost with the possibility of automatic, even cheaper, archival storage, e.g., AWS Glacier. The rest of the architecture is largely the same in the cloud as in the second generation systems, with a downstream data warehouse such as Redshift or Snowflake. This two-tier data lake + warehouse architecture is now dominant in the industry in our experience (used at virtually all Fortune 500 enterprises).

This brings us to the challenges with current data architectures. While the cloud data lake and warehouse architecture is ostensibly cheap due to separate storage (e.g., S3) and compute (e.g., Redshift), a two-tier architecture is highly complex for users. In the first gener-

Data Lakehouse

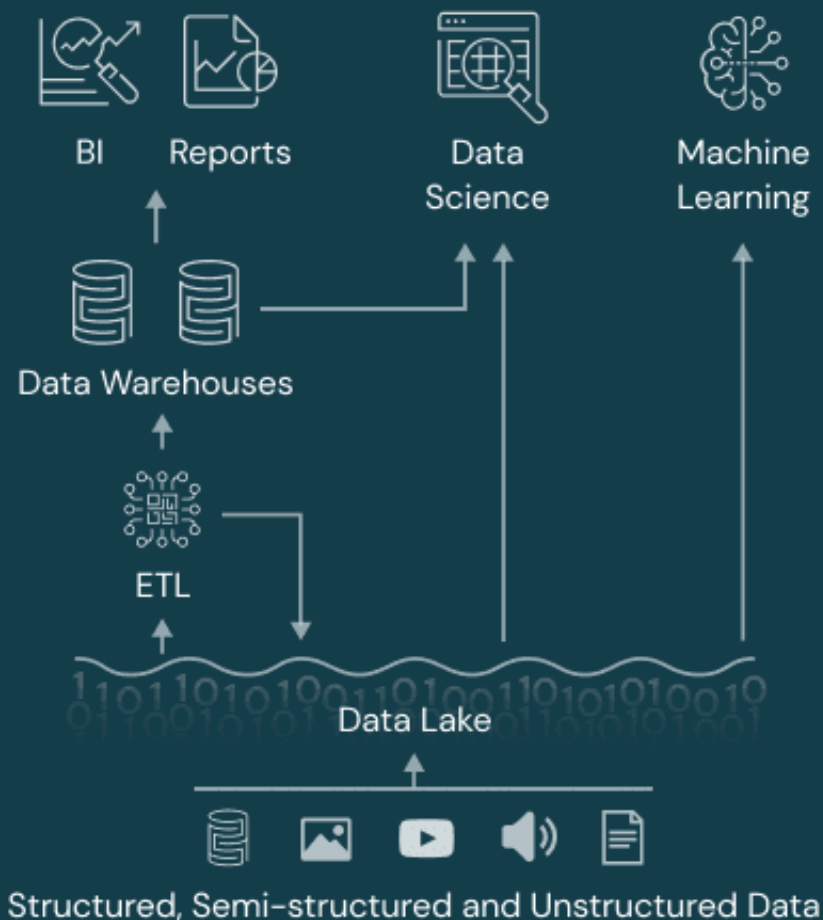
- Goal: DBMS features on top of low-cost, directly-accessible storage
- 1) *Metadata layer* on object store
 - See: Delta Lake, Apache Iceberg
- 2) *Performance optimizations*
 - Separate storage / compute
 - Caching, auxiliary indexes, record ordering
- 3) *ML Interfaces*
 - Map DataFrame API to SQL query plans
 - -> apply SQL optimizer



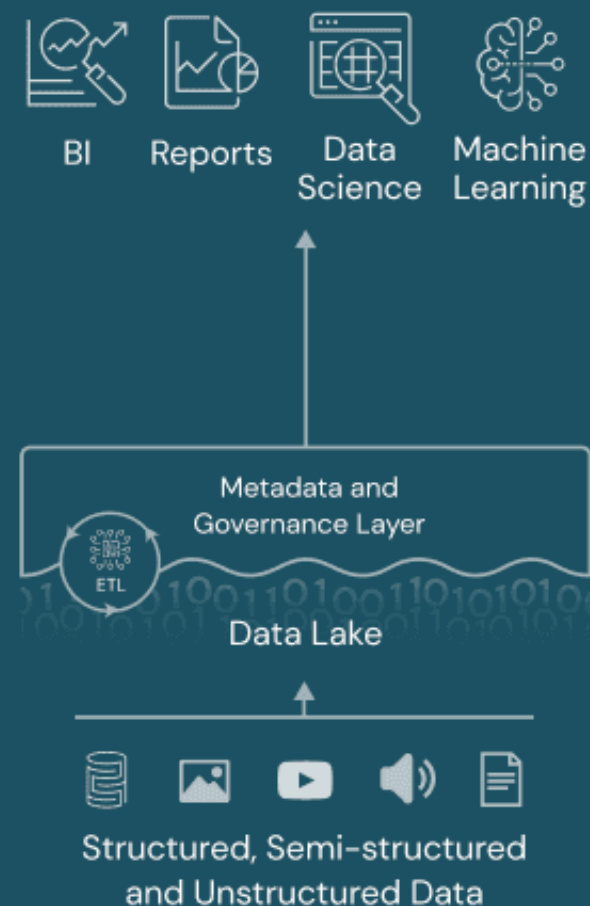
Data Warehouse



Data Lake



Data Lakehouse



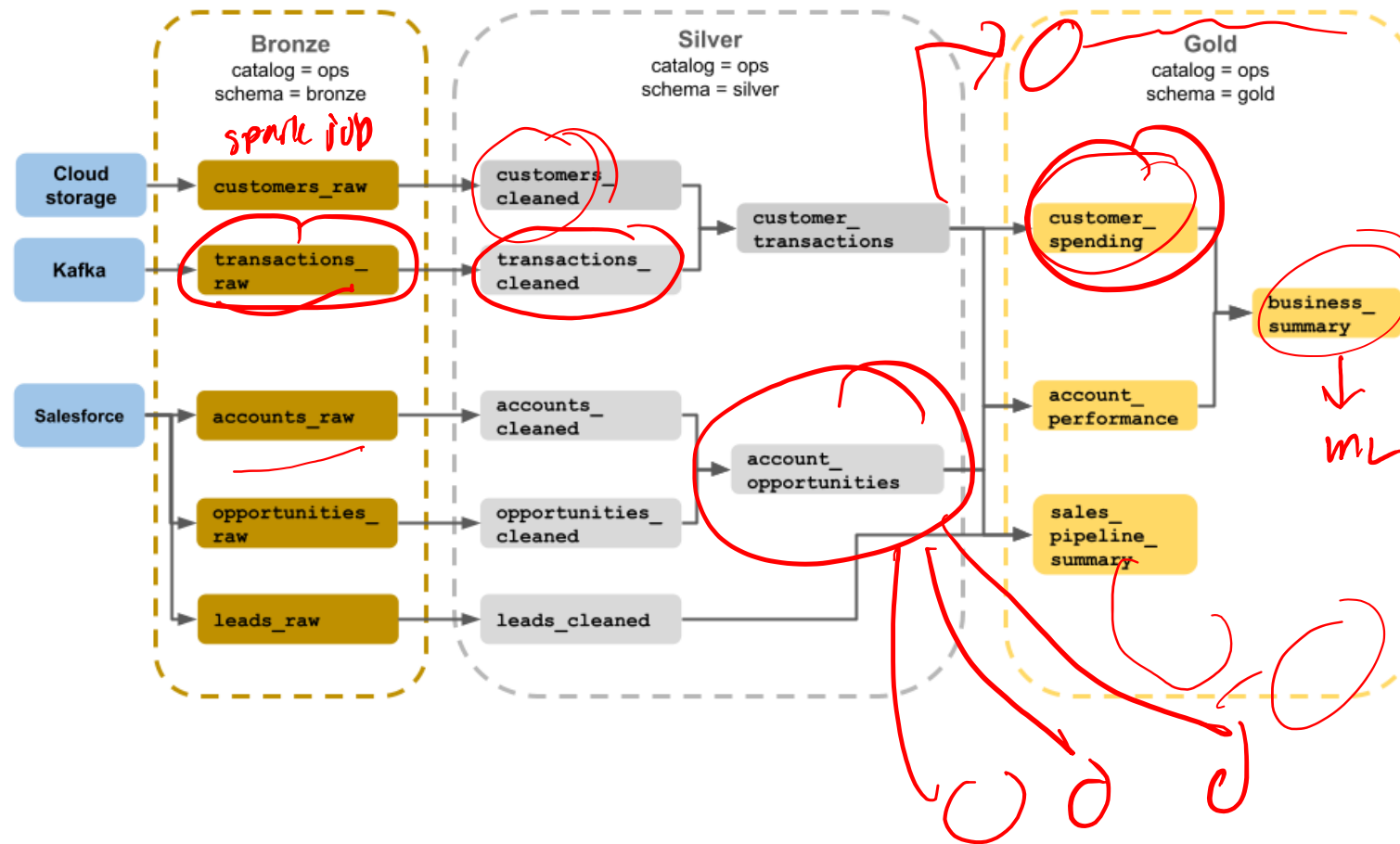
Lake, Warehouse, Lakehouse

	Data lake	Data warehouse	Lakehouse
Schema	On read	On write	On write, enforced by table format
Storage	Object store, open formats	Proprietary, coupled to engine	Object store, open formats
Transactions	None	Full ACID	ACID via metadata layer
Cost profile	Cheapest per TB	Expensive per TB and per query	Lake storage, warehouse semantics
<u>ML fit</u>	Good for raw + unstructured; no guarantees	Poor for unstructured; hard to version	Current default for training data

Pipeline Architecture: ETL vs ELT

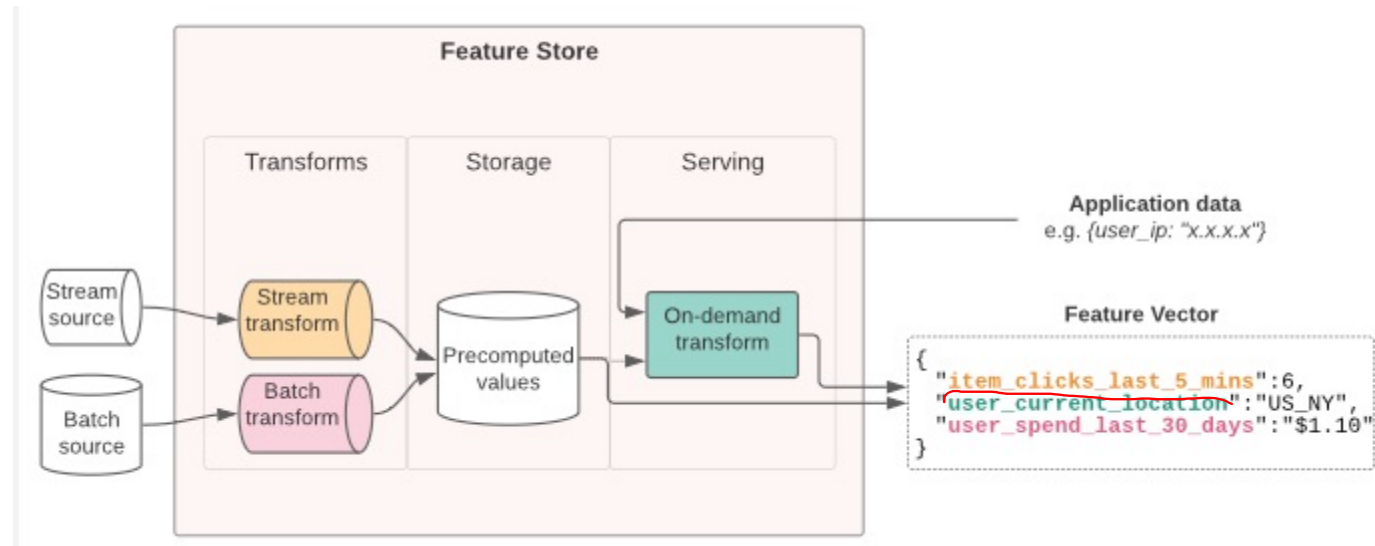
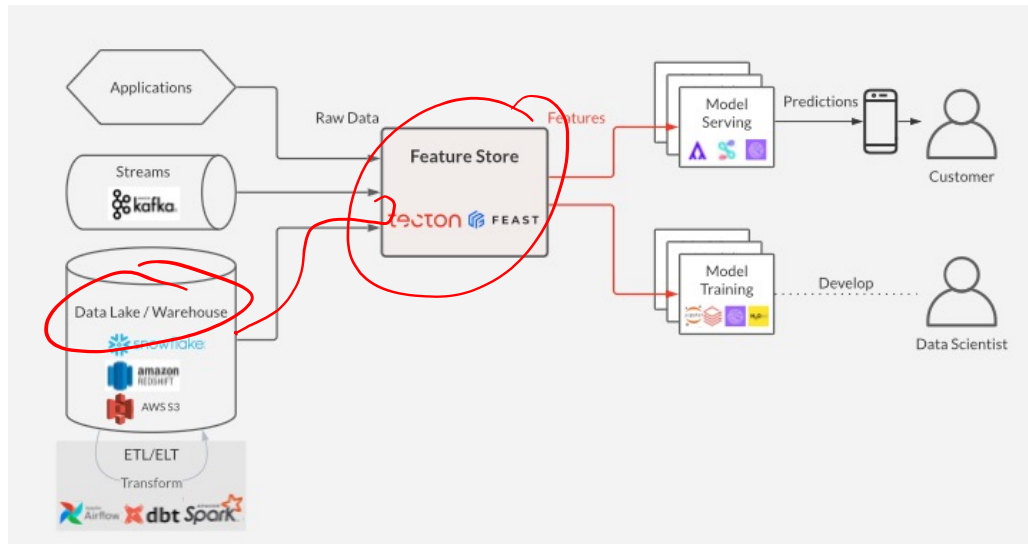
	ETL — extract, transform, load	ELT — extract, load, transform
Transform	Before storage	After storage, on query
What lands	Clean, smaller, opinionated	Raw, larger, uncommitted
Schema change	Expensive — reprocess history	Cheap — schema on read
Governance	Easier: everything is structured	Harder: raw zone needs its own controls
ML use	Feature engineering with settled definitions	Exploration, and training on raw signal
Cost profile	Higher engineering cost up front	Higher storage cost, lower engineering

Medallion Lakehouse Architecture



Feature Stores

- Centralized data platform to store, manage, and serve curated input variables (features) across training and production environments.



Why might we want a feature store?

"Source of truth"

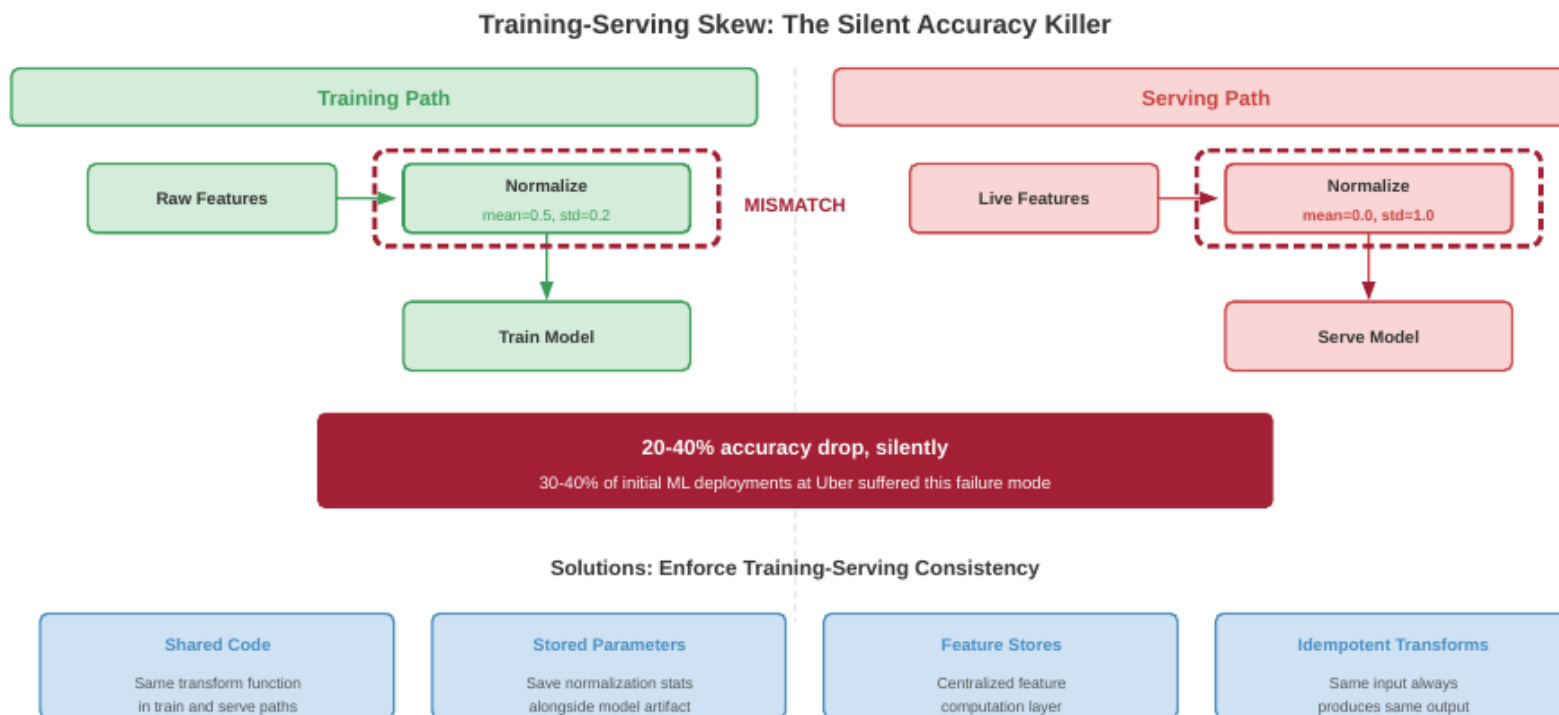
Reduction of computation

Why might we want a feature store?

- Duplicated Effort: Share/reuse feature pipelines (break “data silos”)
- Optimize performance for downstream needs
 - Training: Large “batch” reads -> high throughput
 - Serving: Low-latency, up-to-date values

Why might we want a feature store?

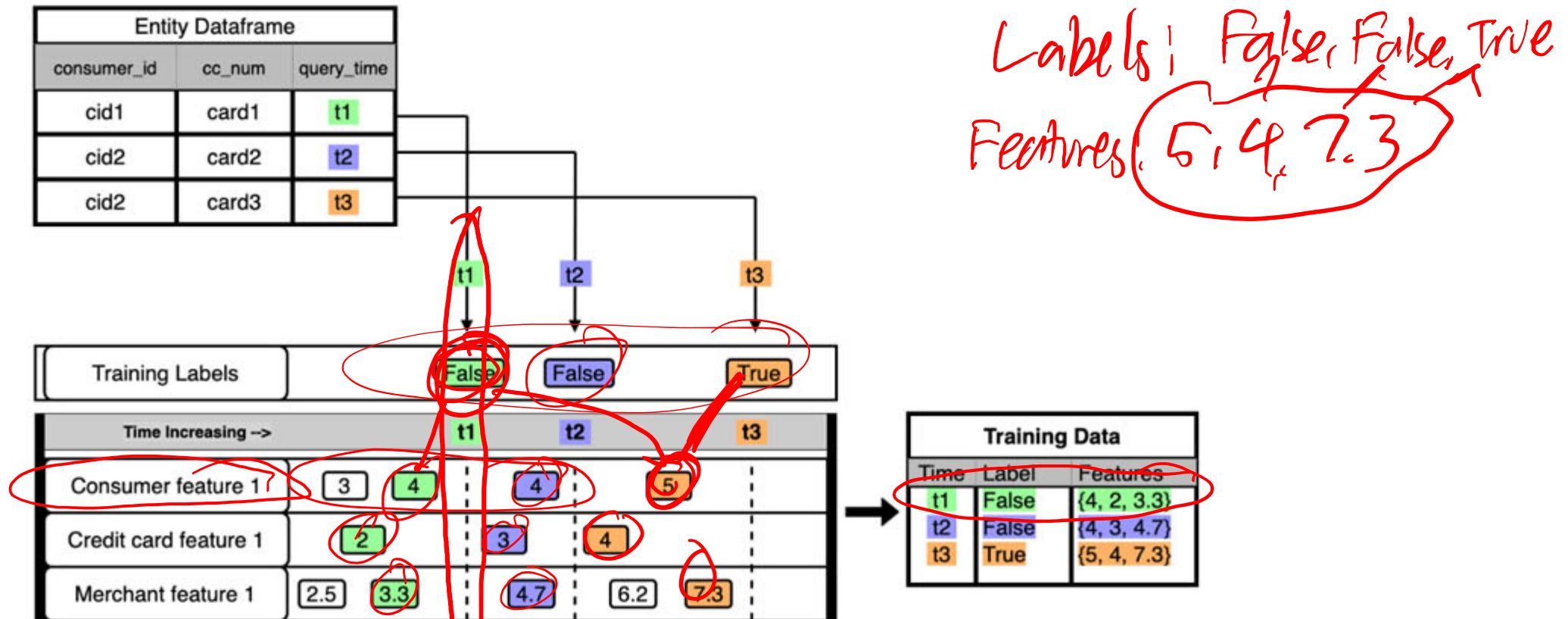
- Eliminate "Training-Serving Skew": Consistent transform logic across training and inference



Consistency is state synchronization, not just code-sharing

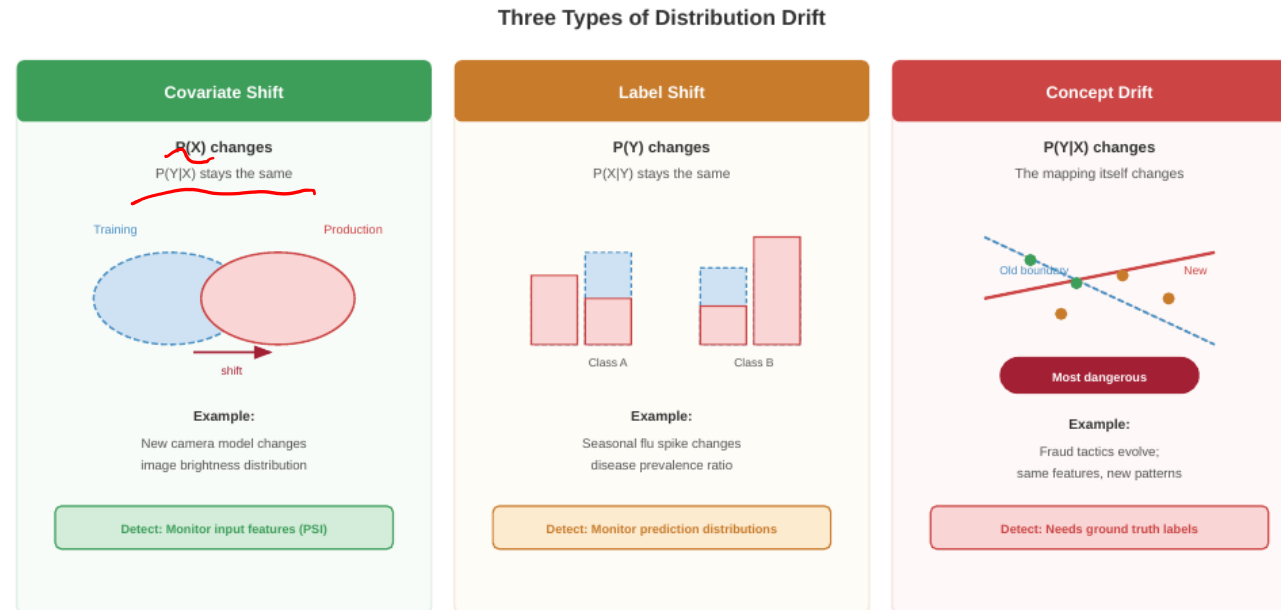
Why might we want a feature store?

- Ensure “point-in-time” correctness
 - To train an accurate model, each training sample has to contain feature values that were present **when the label was generated**



Why might we want a feature store?

- Centralized point to monitor for data drift



Concept drift is delayed because ground truth labels arrive late — the most dangerous silent failure

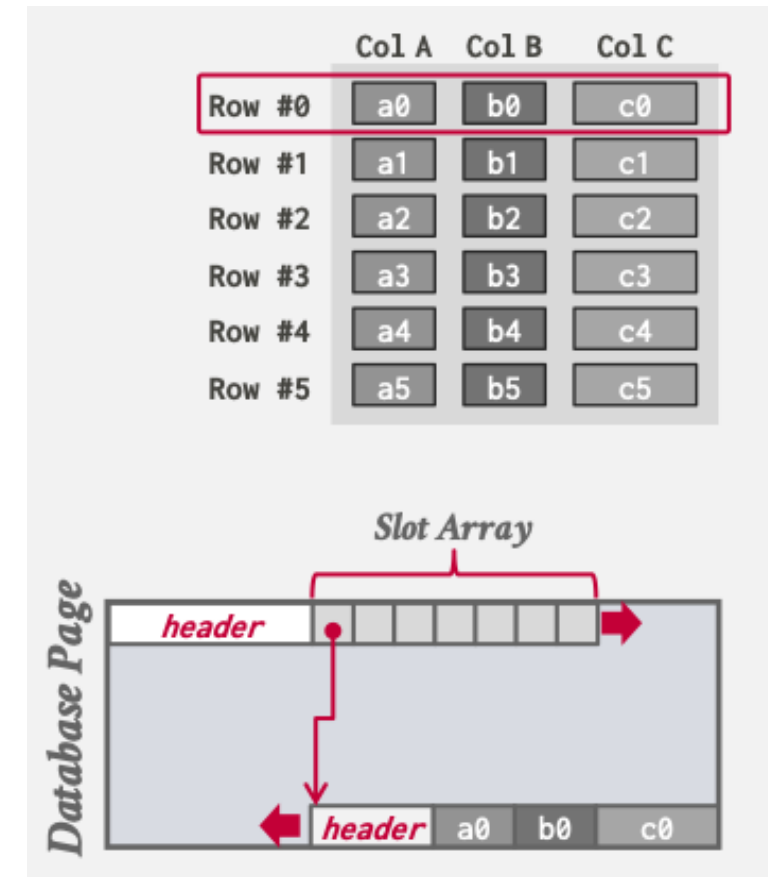
Key: Concept drift is the most dangerous — rules change but detection requires ground truth labels that arrive late.

Storage Models

- Suppose we wanted to store a training dataset with millions of records – e.g. (userID, feature_1, ..., feature_N, label). What file should we store this dataset in?

Storage Models: N-ary Storage Model

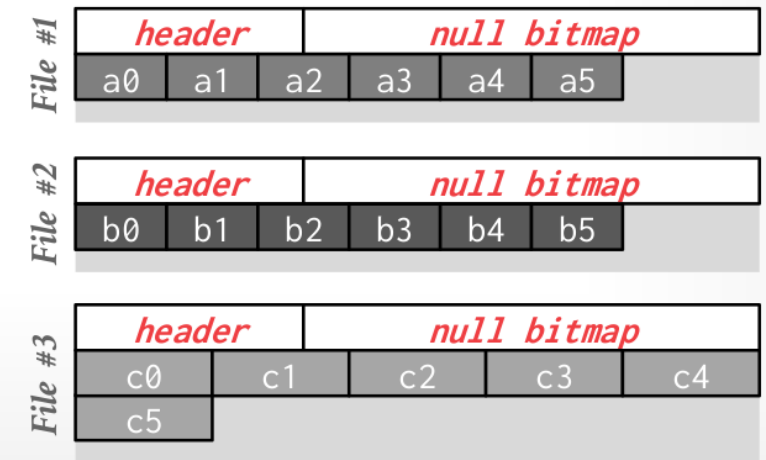
- Row store (n-ary storage model)
 - All attributes for a single record contiguously
- Ideal for
 - Workloads which need random access to individual records
 - e.g. INSERT INTO users VALUES (A,B,C)
 - Workloads which stream the **entire** record



Storage Models (in a nutshell)

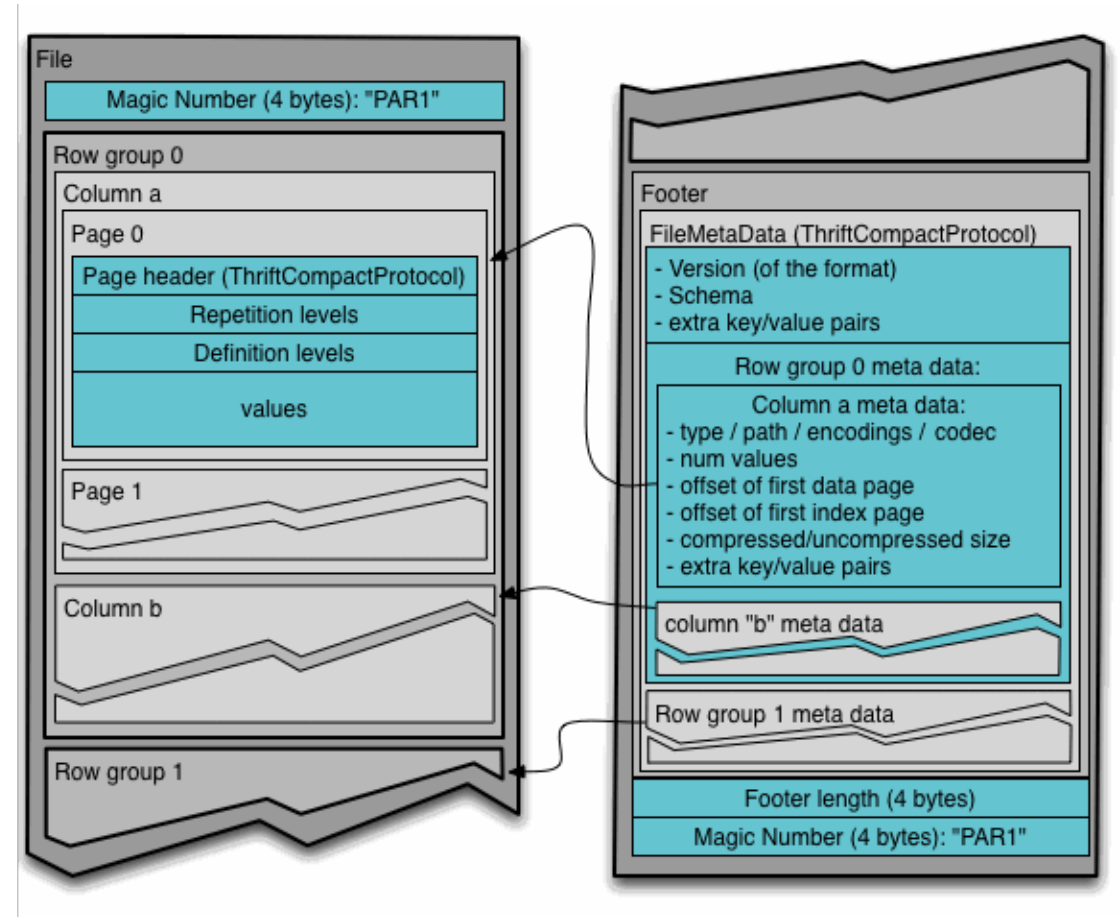
- Column store (Decomposition storage model)
 - Store single attribute for all records contiguously in block
 - Ideal for OLAP workloads which scan subset of table's attributes
 - E.g., `SELECT AVG(salary) FROM employees WHERE department = 'Engineering'`

	Col A	Col B	Col C
Row #0	a0	b0	c0
Row #1	a1	b1	c1
Row #2	a2	b2	c2
Row #3	a3	b3	c3
Row #4	a4	b4	c4
Row #5	a5	b5	c5



Storage Models (in a nutshell)

- Partition Attributes Across (PAX) model
 - Hybrid model
 - Horizontally partition data into **row groups**
 - Vertically partition attributes in each row group into **column chunks**
 - Still gives us row-wise locality when we need to reconstruct rows
- De-facto format for ML: ORC, Parquet, Arrow, Nimble, Vortex



Co-locating Compute and Data



Co-locating Compute and Data

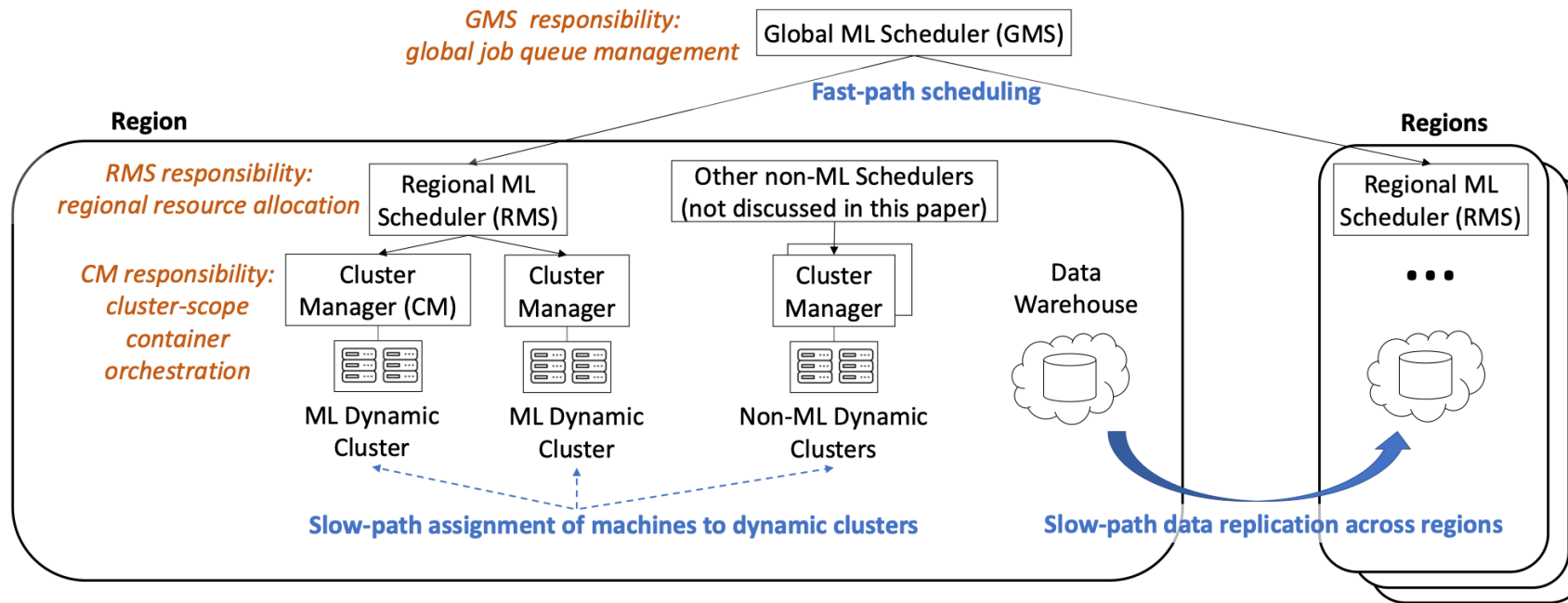
Moving 1 PB over a 10 Gbps link

- Wire time: ~3 weeks.
- Egress cost: ~\$90K at \$0.09/GB every time.
- Engineering time: 6–9 months of pipeline re-validation, schema migration, and lineage rework.

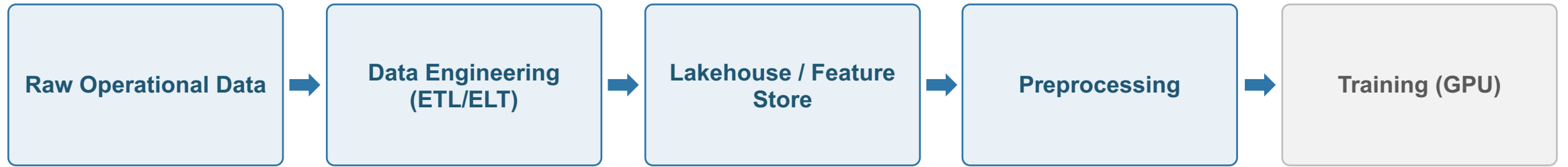
Scale	The right architecture
GB	Move the data to the compute
TB	Co-locate storage and compute
PB	Lakehouse: bring Spark to the storage
PB+, many orgs	Federated / mesh: do not centralize at all

- Storage decisions from the last few slides are hard to reverse — data acquires “gravity” as consumers accumulate around it
- Practical consequence: choose the format, frameworks, and the location before the dataset is large

Example: MAST – Meta’s global ML scheduler



Recap: The Training Data Pipeline



ML Preprocessing

- **"Last-mile" of data processing**
 - f: batch of training samples → tensors in GPU HBM
- **Highly domain-specific, expressed as Python modules**
 - Vision: color augmentation, shear, contrast, ... (e.g. torchvision)
 - Text: tokenize, normalize, pad, embed, ... (e.g. tf.text)
 - Recommendation: hashing, bucketization, sparse feature assembly
- **Unique semantics: random augmentations, data ordering, epoch boundaries**



Unlike “offline” prep, this code lives inside the training job. Written by the modeler, “streaming” (no materialization), and rarely treated as infrastructure.

DataLoaders

```
import os
import pandas as pd
from torchvision.io import decode_image

class CustomImageDataset(Dataset):
    def __init__(self, annotations_file, img_dir, transform=None, target_transform=None):
        self.img_labels = pd.read_csv(annotations_file)
        self.img_dir = img_dir
        self.transform = transform
        self.target_transform = target_transform

    def __len__(self):
        return len(self.img_labels)

    def __getitem__(self, idx):
        img_path = os.path.join(self.img_dir, self.img_labels.iloc[idx, 0])
        image = decode_image(img_path)
        label = self.img_labels.iloc[idx, 1]
        if self.transform:
            image = self.transform(image)
        if self.target_transform:
            label = self.target_transform(label)
        return image, label
```

- **DataLoader abstraction**

- Map-style `__getitem__()`: random access to any sample by index (why is this bad?)
- Iterable-style `__iter__()`: iterable over data samples (why might this be bad?)
- Per-sample transforms, applied lazily

`__getitem__` (or `__iter__()`) is called millions of times per epoch, in Python, and hides storage I/O, decode, and many tensor ops.

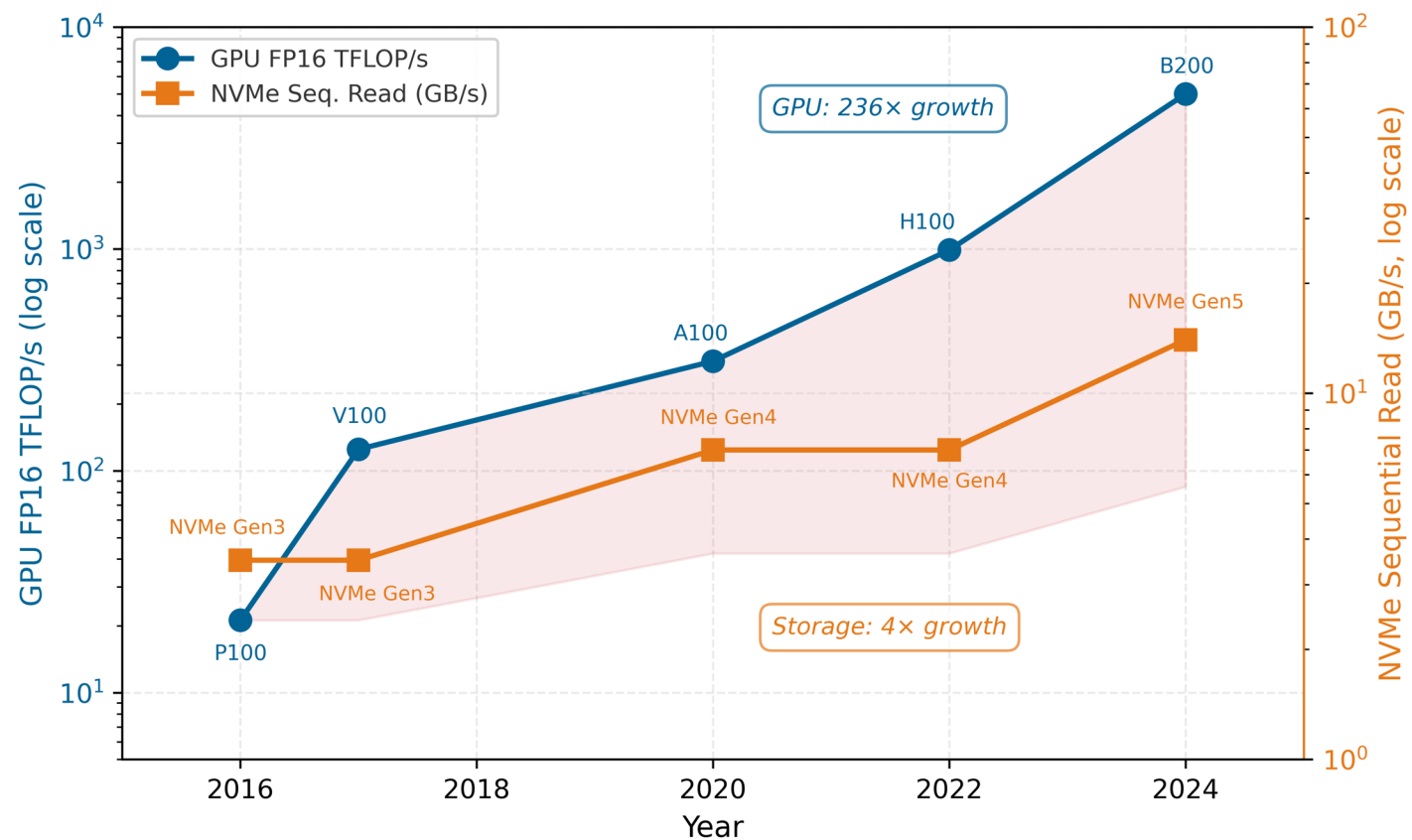
The ML Storage Hierarchy

Tier	Storage	Bandwidth	Cliff vs tier above
0	GPU HBM	3.35 TB/s	—
1	Host DRAM	200 GB/s	17x
2	Local NVMe	7–25 GB/s	10–30x
3	Parallel FS	4 GB/s / node	10x
4	Object storage	~1 GB/s / node	4x

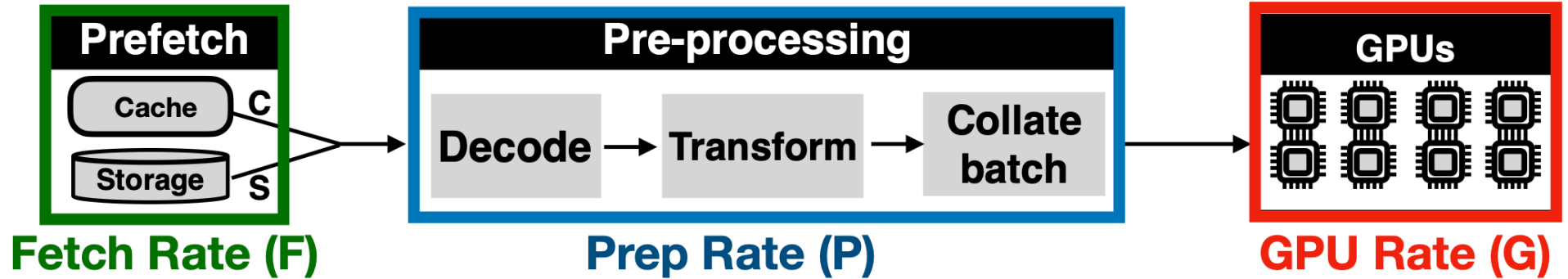
- ~500x gap between GPU HBM and local NVMe!
- HBM can empty in 24ms, but it takes 400ms to refill from DRAM
 - The next batch must *already* be in HBM before the current one finishes.
 - Q: What happens if the batch is not ready yet?

Implication: Each tier is a prefetch buffer for the tier above it. Need to size the storage hierarchy so the entire storage path can feed the tier above it.

The I/O Wall



Data Stalls



A data stall occurs if $G > \min(F, P)$

- F — fetch rate: reading bytes from cache or storage into host memory
- P — prep rate: decode, transform, collate into a batched tensor
- G — GPU rate: how fast the accelerator can consume batches
- **The input pipeline is a throughput system, so either F or P can starve the GPU.**

The Pipeline Equation

$$\text{BW_required} = \text{N_GPUs} \times \text{U_target} \times (\text{S_batch} / \text{T_iteration})$$

Variable	Meaning	ImageNet on 256 GPUs
N_GPUs	Number of accelerators	256
U_target	Target utilization (0.8–0.95)	0.80
S_batch	Local batch size in bytes	256 × 150 KB
T_iteration	Forward–backward time	0.2 s
BW_required	What the data pipeline must sustain	39.3 GB/s, continuously throughout the training job

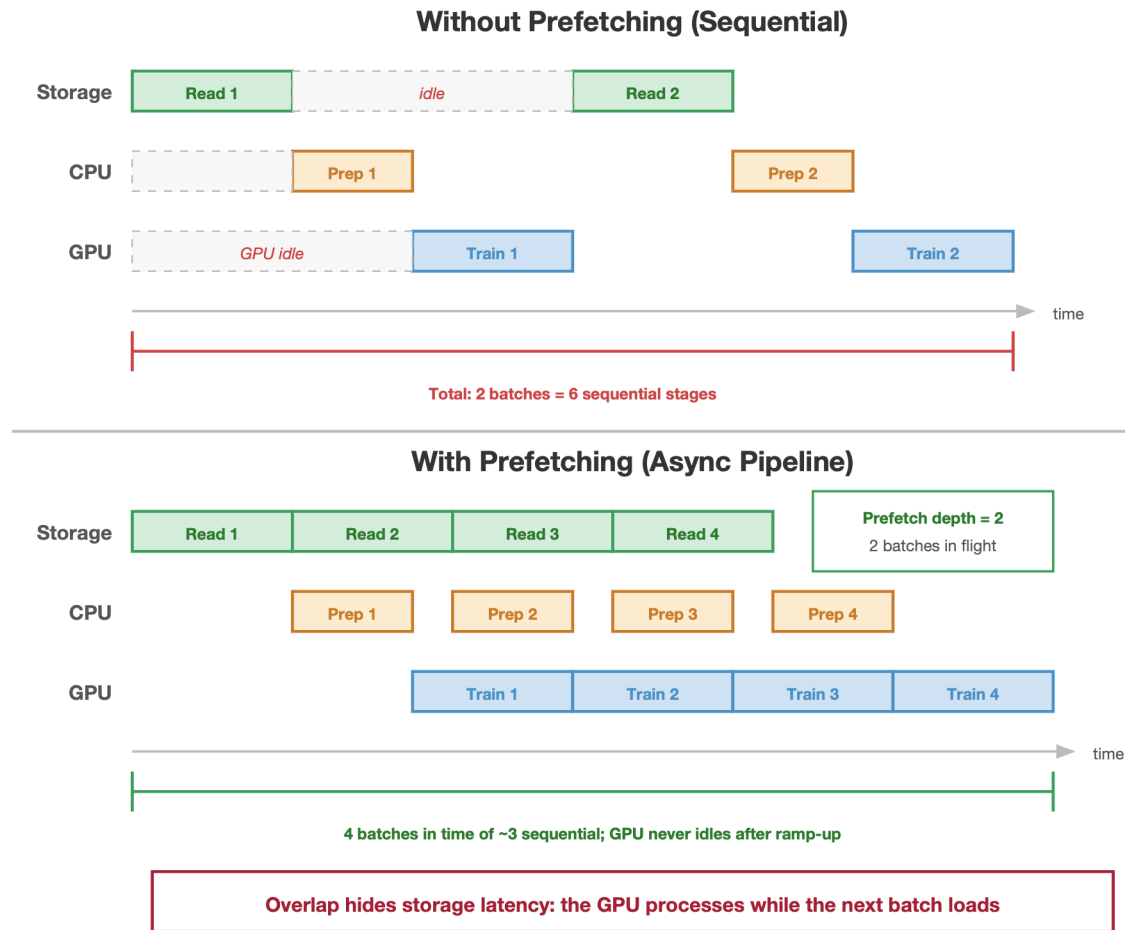
Data Stall Ratio

$$\text{Data stall \%} = (T_{\text{step}} - T_{\text{compute}}) / T_{\text{step}} \times 100$$

Stall ratio	Diagnosis	Action
< 2%	Healthy pipeline	Monitor. Do not over-invest here
2–10%	Marginal — will break at scale	Audit the pipeline, add prefetch depth
> 10%	Clear storage bottleneck	Redesign the data path

- **A 5% stall on 1,000 GPUs at \$2/GPU-hour wastes about \$16,800 in a single week. 1% is massive at scale.**
- **Consequence: Upgrading A100 to H100 without touching storage increases the stall ratio. Need to re-think storage architecture alongside accelerator upgrades.**

Mitigation 1: Pipelining and Prefetching



- Example:
 - $T_{\text{compute}} = 200\text{ms}$
 - $T_{\text{I/O}} = 250\text{ms}$
- Stall% without pipelining?
- Stall% with?

Mitigation 1: Pipelining and Prefetching

$$Q_{\min} = \text{ceil}(T_{\text{io,p99}} / T_{\text{compute}})$$

Given	Value	Prefetch depth needed
T_compute per batch	200 ms	—
T_io average	250 ms	ceil(250/200) = 2 batches
T_io p99 (tail)	500 ms	ceil(500/200) = 3 batches

- **Size the buffer for P99, not the average**
 - Object storage makes this worse: P99 latencies of 500+ ms are routine
- The cost of depth is host memory. Q_min batches of tensors must compete with everything else on the node.

Mitigation 2: Scale-up within the node

Multi-process workers

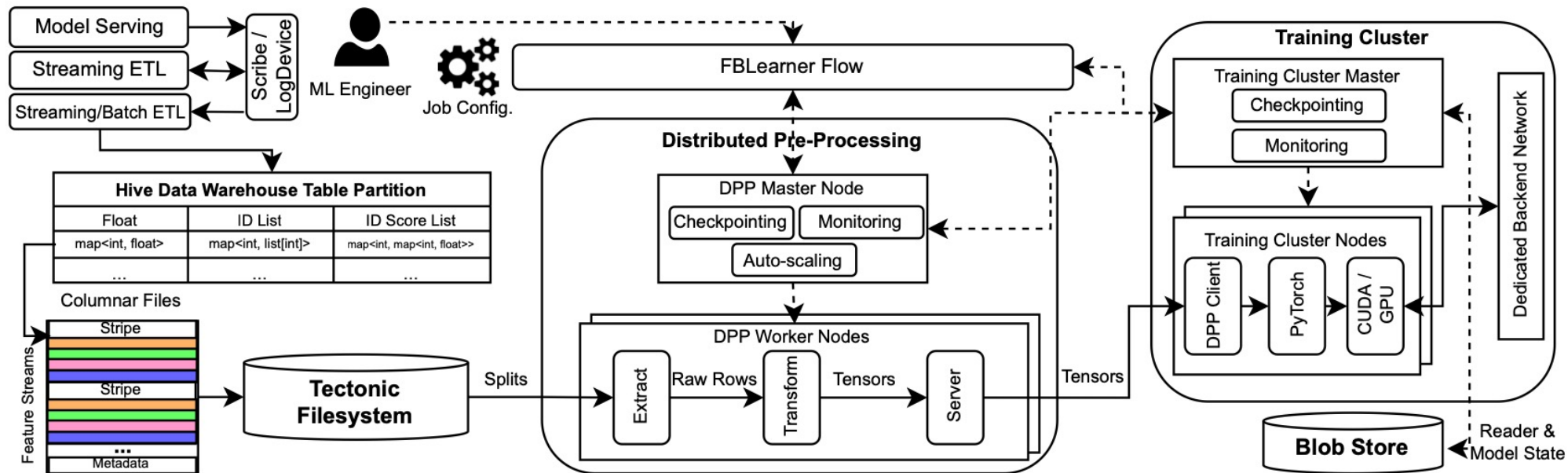
Use multiple Python worker processes (`num_workers`) to overlap loading with compute and sidestep the GIL. Helps when P is the bottleneck. Costs host memory and CPU contention.

Pinned memory

Land tensors in physical, non-swappable pages so the GPU can DMA directly. Helps when P is the bottleneck by removing a CPU-side copy.

- How to diagnose if G, F or P is the bottleneck?
- Try these first, easy configuration change without much cost
- Downside of these options?

Mitigation 3: Scale-out and disaggregate



- Run preprocessing (and storage) as a dedicated, independently scaled service. Ship finished tensors to the trainer
 - Helps when P (disagg compute) or F (disagg storage) is the bottleneck, and breaks the fixed disk:CPU:GPU ratio of the training node

Mitigation 3: Change the graph

Caching / materialization

Write (partially) preprocessed data back to storage and reuse it across epochs and jobs. Helps P, and helps F when the materialized form is smaller. Constraint: you cannot materialize past a random augmentation.

Storage optimizations

Change placement (remote → local SSD, or into page cache) or format (row → columnar, or large sequential shards). Helps F.

Operator fusion/kernels

Combine adjacent operators to avoid materializing intermediates and keep data in cache. Helps P.

Reordering

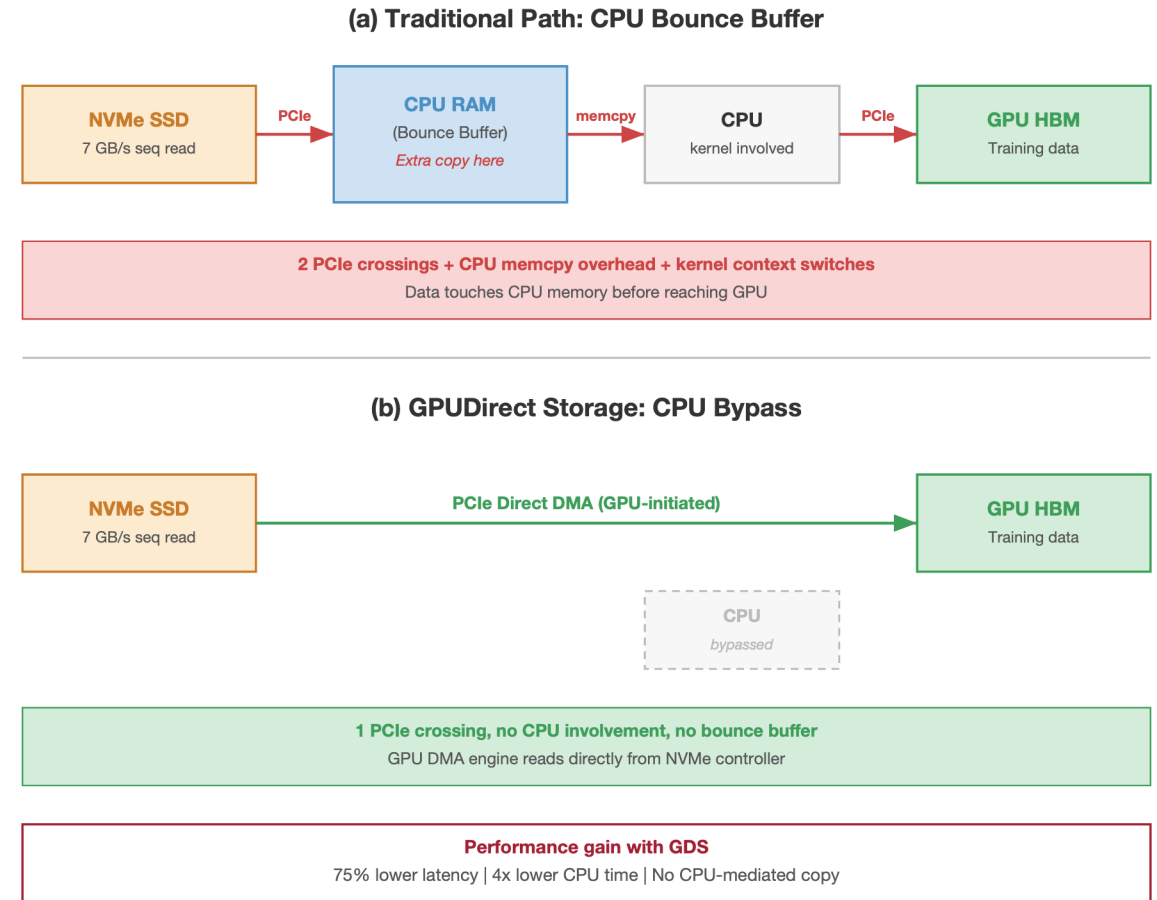
Change operator order to do less work, without changing semantics. Helps P.

Mitigation 3: Change the graph

- Question: When should we run a transformation “offline” versus “online”?

Mitigation 4: Offloading

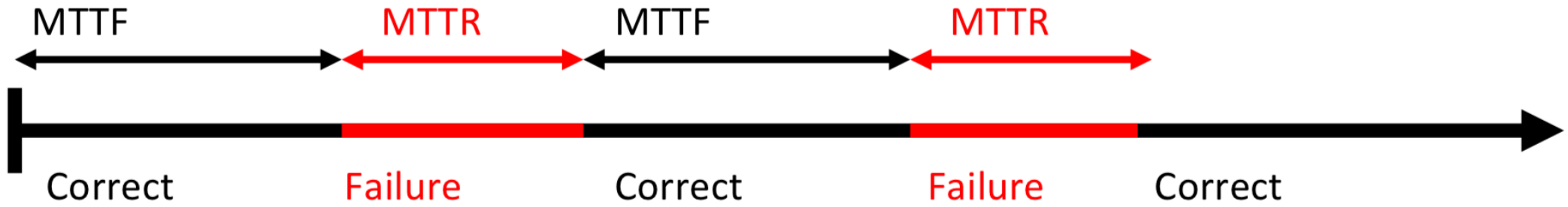
- Use a different implementation, or different hardware, for the same operator
- Vectorized CPU kernels (AVX), GPU decode and augmentation (e.g. DALI, nvJPEG), dedicated decoders.
- Helps P — but can compete for GPU resources
- Offloading augmentation to the GPU relieves the CPU and consumes GPU cycles. It works only when the GPU has spare cycles.



Summary

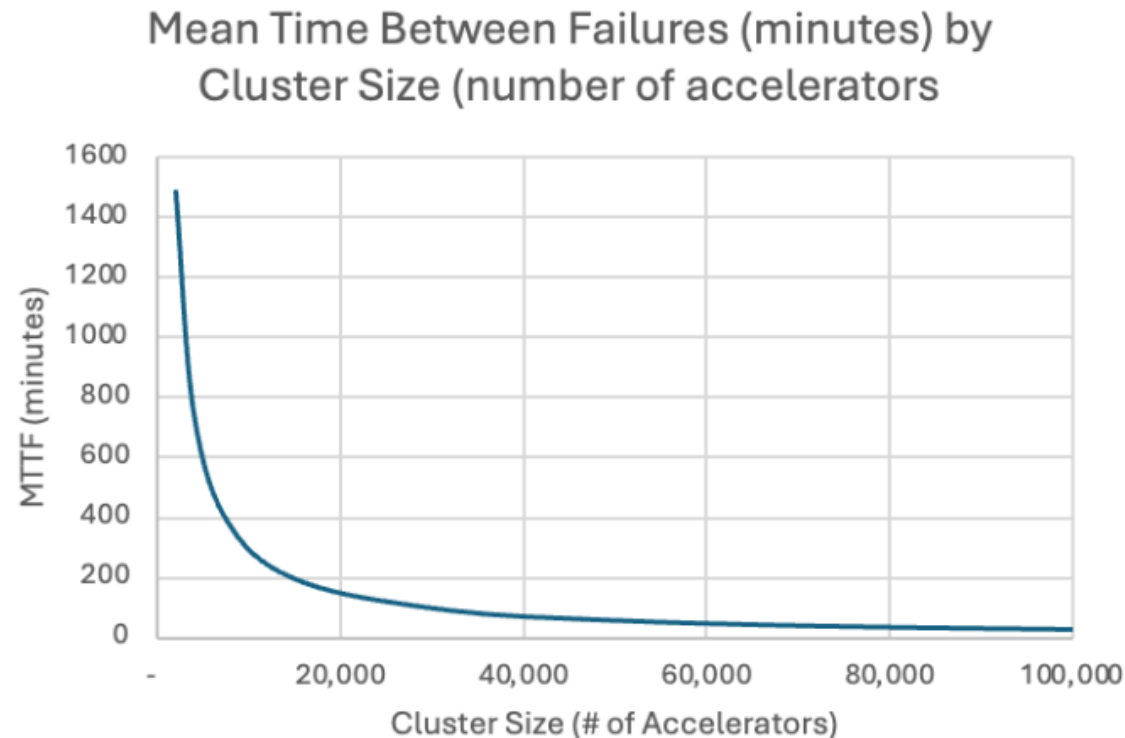
- Measure first: Figure out if G, P, or F is your bottleneck
- Move work in space
 - Scale-up, scale-out, offload, use GDS
- Move work in time
 - Cache/materialize intermediate values, prefetch
- Move work in order
 - Modify your graph to reduce work – move ops between online/offline, fuse, or reorder ops

Failure at Scale



Failure at Scale

- Rare events individually -> common at scale
- Assume MTTF of a GPU is 3 years...
 - 1000 GPUs -> MTTF of one GPU is ~1 day
 - 10000 GPUs -> MTTF ~3 hours
 - 100000 GPUs -> MTTF ~**15 minutes**



Handling Failures

- Traditional datacenter applications
 - **Tolerating Faults:** Add layers of redundancy to automatically detect and repair faults (e.g., replicas, RAID, ECCs)
 - **Graceful Degradation:** Allow faults to be visible, but maintain some level of functionality (e.g., 360p instead of 4K under heavy load)
- ML training
 - **Synchronous training:** *Any* failure will cause the *entire* training job to fail

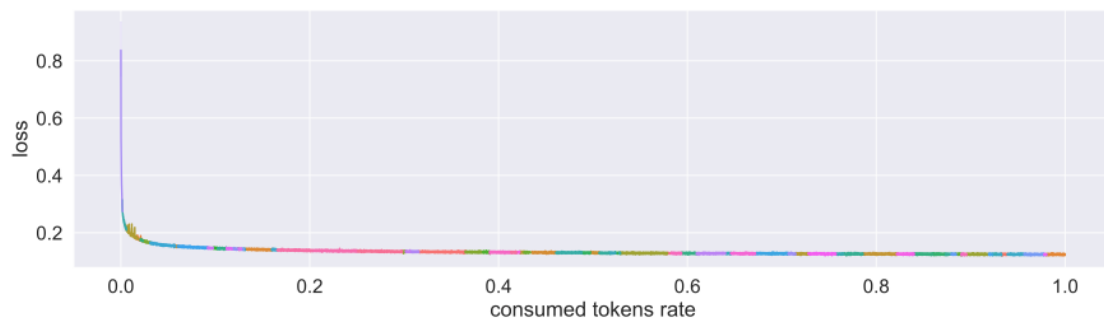
Real-world Examples

MegaScale: Scaling Large Language Model Training to More Than 10,000 GPUs

Ziheng Jiang^{1,*} Haibin Lin^{1,*} Yinmin Zhong^{2,*} Qi Huang¹ Yangrui Chen¹ Zhi Zhang¹
Yanghua Peng¹ Xiang Li¹ Cong Xie¹ Shibiao Nong¹ Yulu Jia¹ Sun He¹ Hongmin Chen¹
Zhihao Bai¹ Qi Hou¹ Shipeng Yan¹ Ding Zhou¹ Yiyao Sheng¹ Zhuo Jiang¹
Haohan Xu¹ Haoran Wei¹ Zhang Zhang¹ Pengfei Nie¹ Leqi Zou¹ Sida Zhao¹
Liang Xiang¹ Zherui Liu¹ Zhe Li¹ Xiaoying Jia¹ Jianxi Ye¹ Xin Jin^{2,†} Xin Liu^{1,†}

¹ByteDance ²Peking University

Training an LLM with one trillion tokens can take weeks. The scale and time are orders of magnitude larger than those of regular DNN training jobs. Failures and stragglers are the norm rather than the exception for LLM training. At such a scale, the consequences of failures and stragglers are devastating. Failures are very expensive, and it is critical to reduce the recovery time, given the large scale. A straggler not only affects its own work, but slows down the entire job involving tens of thousands of GPUs.



Meta

The Llama 3 Herd of Models

Llama Team, AI @ Meta¹

¹A detailed contributor list can be found in the appendix of this paper.

Modern artificial intelligence (AI) systems are powered by foundation models. This paper presents a new set of foundation models, called Llama 3. It is a herd of language models that natively support multilinguality, coding, reasoning, and tool usage. Our largest model is a dense Transformer with 405B parameters and a context window of up to 128K tokens. This paper presents an extensive empirical evaluation of Llama 3. We find that Llama 3 delivers comparable quality to leading language models such as GPT-4 on a plethora of tasks. We publicly release Llama 3, including pre-trained and post-trained versions of the 405B parameter language model and our Llama Guard 3 model for input and output safety. The paper also presents the results of experiments in which we integrate image, video, and speech capabilities into Llama 3 via a compositional approach. We observe this approach performs competitively with the state-of-the-art on image, video, and speech recognition tasks. The resulting models are not yet being broadly released as they are still under development.

Date: July 23, 2024

Website: <https://llama.meta.com/>

During a 54-day snapshot period of pre-training, we experienced a total of 466 job interruptions.

Component	Category	Interruption Count	% of Interruptions
Faulty GPU	GPU	148	30.1%
GPU HBM3 Memory	GPU	72	17.2%
Software Bug	Dependency	54	12.9%
Network Switch/Cable	Network	35	8.4%
Host Maintenance	Unplanned Maintenance	32	7.6%
GPU SRAM Memory	GPU	19	4.5%
GPU System Processor	GPU	17	4.1%
NIC	Host	7	1.7%
NCCL Watchdog Timeouts	Unknown	7	1.7%
Silent Data Corruption	GPU	6	1.4%
GPU Thermal Interface + Sensor	GPU	6	1.4%
SSD	Host	3	0.7%
Power Supply	Host	3	0.7%
Server Chassis	Host	2	0.5%
IO Expansion Board	Host	2	0.5%
Dependency	Dependency	2	0.5%
CPU	Host	2	0.5%
System Memory	Host	2	0.5%

Table 5 Root-cause categorization of unexpected interruptions during a 54-day period of Llama 3 405B pre-training. About 78% of unexpected interruptions were attributed to confirmed or suspected hardware issues.

Handling ML Training Failures

- Detect failures
 - Static errors (e.g., crashes): Watchdogs, heartbeats, ...
 - Transient errors (e.g., bitflips): Checksums, ECCs, ...
 - Data errors: Loss monitoring, data distribution checks, validation curve, ...
- Repair failures
 - Swap out failed node with replacement on hot standby
- Resume
 - Restart the training job

Maximizing ML “Goodput”

- Failures
 - Minimize progress lost: Take frequent checkpoints and resume on failure
- Recovery
 - Minimize MTTR: Repair fast (hot standby) and load data from “near” storage
- Checkpointing
 - Minimize the time to take checkpoint: Async snapshots



Checkpointing

- Model state:
 - Weights (4B), Optimizer State (Adam: 4B moment + 4B variance)
- Learning rate schedulers
- Data loader state
- Hyperparameters
- ...

Checkpoint Requirements

- Checkpoint size: ~12B / parameter
- Write: checkpoint size (one data parallel replica)
 - ~12 TB for a 1T parameter model
 - Bursty: at every checkpoint
 - Once every 10 minutes -> **~2 PB *per day***
- Read: checkpoint size * DP degree
 - e.g., DP = 128
 - ~12 TB * 128 replicas = **~1.5 PB to load model**

Changing Storage Bottlenecks

175B-parameter model, 30-day run, 256 nodes	Volume
Training dataset (compressed)	3 TB
Model weights (FP16)	350 GB
Optimizer state (FP32)	1,400 GB
A single full checkpoint	1,750 GB
All checkpoints, 30 days at 10-minute intervals	~7.6 PB

The inversion: checkpoint I/O (~7.6 PB written) is much larger than training-data reads (3 TB read). For frontier-scale models*, checkpointing, not data loading, is the dominant storage workload.

- It competes for the same bandwidth your input pipeline needs
- It is also bursty: every worker writes simultaneously, so the file system must be provisioned for peak, not average
 - The standard mitigation is staging — write to local NVMe fast, replicate to the parallel FS in the background.

*LLMs, the story is different for models which train on non-text inputs which require large datasets (e.g., images, videos)